

Nina Corvelo Benz

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<https://ninacobe.github.io>

PROFILE SUMMARY

My **research interests** lie in the design of **collaborative decision-support systems** to improve human-decision making, leveraging tools from **machine learning, causality** and **uncertainty estimation**.

EDUCATION	Ph.D. in Computer Science 06/2021–10/2025 Max Planck Institute for Software Systems (MPI-SWS) & ETH Zurich (ELLIS PhD Program) <i>Thesis:</i> Counterfactual Reasoning and Uncertainty Quantification for AI-Assisted Decision Making <i>Advisor:</i> Manuel Gomez-Rodriguez Spending autumn/winter 2024 at MPI of Biochemistry, Munich – hosted by Karsten Borgwardt
	Master of Science in Computer Science 09/2018–08/2020 ETH Zurich, Zurich, Switzerland <i>Grade: 5.36/6</i> <i>Focus:</i> Theoretical Computer Science and Machine Learning <i>Thesis:</i> Using Inhibitory Signals for Error Encodings in Networks with Tuning Curves <i>Advisor:</i> Angelika Steger
	Bachelor of Science in Computer Science 09/2014–09/2018 ETH Zurich, Zurich, Switzerland <i>Grade: 5.42/6</i> <i>Thesis:</i> Call Admission Problems on Trees <i>Advisor:</i> Juraj Hromkovič Spent spring 2017 at POSTECH, Pohang, South Korea – exchange program
PROFESSIONAL EXPERIENCE	Postdoctoral Researcher 12/2025 - Now Max Planck Institute of Biochemistry, Munich, Germany
	Doctoral Researcher 06/2021 - 11/2025 Max Planck Institute for Software Systems, Kaiserslautern, Germany
	Research Intern 06/2020–11/2020 Oracle Labs Switzerland, Zurich, Switzerland <i>Topics:</i> Graph Structure based Entity Resolution, ML boosted Address Matching
PUBLICATIONS	Conference Proceedings
(*): equal contribution (A): alphabetical ordering	Conference publications in computer science are archival and going through highly selective peer review. Evaluation of Large Language Models via Coupled Token Generation <i>AISTATS, 2026</i> Nina Corvelo Benz, Stratis Tsirtsis, Eleni Straitouri, Ivi Chatzi, Ander Artoola Velasco, Suhas Thejaswi, Manuel Gomez-Rodriguez

2. **Human-Alignment Influences the Utility of AI-assisted Decision Making**
Scientific Reports 15.1, 2025
Nina Corvelo Benz and Manuel Gomez-Rodriguez
3. **Towards Human-AI Complementarity in Matching Tasks**
The Third Workshop on Hybrid Human-Machine Learning and Decision Making (HLDM'25) at ECML-PKDD 2025
 Adrian Arnaiz-Rodriguez, **Nina Corvelo Benz**, Suhas Thejaswi, Nuria Oliver, and Manuel Gomez-Rodriguez
4. **Counterfactual Token Generation in Large Language Models**
4th Conference on Causal Learning and Reasoning (CLear 2025)
 Ivi Chatzi, **Nina Corvelo Benz**, Eleni Straitouri, Stratis Tsirtsis, Manuel Gomez-Rodriguez
5. **Matchings, Predictions and Counterfactual Harm in Refugee Resettlement Processes**
4th Conference on Causal Learning and Reasoning (CLear 2025)
 Seungeon Lee, **Nina Corvelo Benz**, Suhas Thejaswi, Manuel Gomez-Rodriguez
6. **Detecting antimicrobial resistance through MALDI-TOF mass spectrometry with statistical guarantees using conformal prediction**
29th Conference on Research in Computational Molecular Biology (RECOMB 2025)
Nina Corvelo Benz*, Lucas Miranda*, Dexiong Chen, Karsten Borgwardt, Janko Sattler
7. **Human-Aligned Calibration for AI-Assisted Decision Making**
37th Conference on Neural Information Processing Systems (NeurIPS 2023)
Nina Corvelo Benz and Manuel Gomez Rodriguez
8. **Counterfactual Inference of Second Opinions**
38th Conference on Uncertainty in Artificial Intelligence (UAI 2022)
Nina Corvelo Benz and Manuel Gomez Rodriguez
9. **Call Admission Problems on Trees with Advice**
30th International Workshop on Combinatorial Algorithms (IWOCA 2019)
 (A) Hans-Joachim Bröckenhauer, **Nina Corvelo Benz** and Dennis Komm

Pre-prints

1. **Learning to Decide with AI Assistance under Human-Alignment**
In Arxiv, 2026.
Nina Corvelo Benz, Eleni Straitouri, Manuel Gomez-Rodriguez
2. **Tokenization Multiplicity Leads to Arbitrary Price Variation in LLM-as-a-service**
In Arxiv, 2025.
 Ivi Chatzi, **Nina Corvelo Benz**, Stratis Tsirtsis, Manuel Gomez-Rodriguez

TEACHING

Teaching Assistant

Winter Semester 2023

University of Kaiserslautern-Landau (RPTU), Kaiserslautern, Germany

Seminar: Human-Centric Machine Learning

Teaching Assistant

Winter Semester 2018 & 2019

ETH Zurich, Zurich, Switzerland

Course: Theoretical Computer Science

GRANTS AND AWARDS	Maximize Incubator Grant 2025
	Funding of approximately 50,000 Euros awarded for the development of an accessible virtual training coach
	Max Planck Career Evolution Games Best Business Pitch Award 2024
SCIENTIFIC COMMUNITY ACTIVITIES	Workshop organizing
	NeurIPS 2026 Workshop Chair Assistant 2026
	NeurIPS 2025 Workshop Chair Assistant 2025
	ICML Workshop on Counterfactuals in Minds and Machines, Hawaii, USA 07/2023
	Co-organized with Stratis Tsirtsis, Ricardo Dominguez-Olmedo, Isabel Valera, Manuel Gomez-Rodriguez, Amir-Hossein Karimi, Thorsten Joachims and Sarah Wu
	Reviewing
	NeurIPS 2025, AISTATS (2023, 2026), TLMR (2024, 2025), ICML 2024, UAI (2023, 2024, 2025)
TECHNICAL SKILLS	Programming Languages: Python, Java, C++, R
	Database Systems: MySQL
	Libraries: Numpy, Pandas, Keras, PGX, StellarGraph, Elasticsearch
	Editing tools: L ^A T _E X, Microsoft Excel, MS Word, MS PowerPoint
LANGUAGES	English (fluent), German (fluent), Portuguese (native), Spanish (advanced), Greek (Beginner), Korean (Beginner)